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BE (8th Semester)

Examination, Nov.-Dec., 2013

Branch : Mechanical

PRODUCTION MANAGEMENT

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) is compulsory.

(ii) Attempt any two parts from (b), (c) and (d) in each question.

Q. 1. (a) Define production management and write its objectives. 2

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(b) Explain in brief the different types of production system. Mention the factors which affect the success of production function. 7

(c) Explain with a graph the various stages of product life-cycle. 7

(d) ABC company is investigating the decision whether to make or buy a plastic packaging which is currently being purchased at Rs. 7 each. The demand estimates are shown below.

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Demand (units)	20,000	30,000	40,000	50,000	60,000
Chance (%)	10	30	40	15	5

The decision to manufacture in-house costs the company an annual fixed cost of Rs. 80,000/- toward renovation and conditioning variable costs are estimated at Rs. 5 per unit. Give your decision whether to make or buy. 7

Q. 2. (a) What is sales forecasting ? Mention the forecasting methods. 2

(b) What is time series analysis ? What are the components of time series ? How the forecast is made from time series ? 7

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(c) What are seasonal variations ? Why are they important for sales forecast ? 7

(d) The past data regarding the sales of SPMS for the last five years is given. Using the least square method, fit a straight line. Estimate the sales for the year 1996 and 1997. 7

Year	1991	1992	1993	1994	1995
Sales ('00)	35	56	79	80	40

Q. 3. (a) Define production planning and control. Mention its need. 2

(b) What are the functions of production planning and control. Explain Centralised and Decentralised PPC. 7

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(c) There are seven jobs which are to be pressed first on Machine I and then on Machine II. Processing time in hours are given below.

Job	A	B	C	D	E	F	G
Machine I	6	24	30	12	20	22	18
Machine II	16	20	20	13	24	2	6

Find the optimal sequence and the total elapsed time. Compute the idle time on Machine II. 7

(d) Define material handling. Explain the principles of material handling. 7

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Q. 4. (a) What are the objectives of mate management ?

(b) State the objectives of purchasing. Explai techniques used for making purchases.

(c) What are the objectives of store keepi Compare closed store and open store.

(d) Describe ABC analysis. State applications.

Q. 5. (a) Define Quality Control. What are the typ control charts ?

(b) What is the meaning of Quality Conformance ? Explain the factors w influence Quality of Conformance.

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(c) What do you mean by process control ? State the objectives of $\bar{X}$ and R charts. 7

(d) What is Acceptance Sampling ? Write down the advantages and limitations. 7

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